Contribution Write-Up

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For this project, I focused mainly on the modeling and risk score development, which ended up being one of the most technically challenging parts of the process. Once the data was cleaned and ready, I built out all three classification models: Logistic Regression, Decision Tree, and Random Forest, and ran detailed comparisons across metrics like accuracy, precision, recall, and F1 score. One of the biggest takeaways from this phase was realizing how misleading accuracy can be, especially with imbalanced data. While Logistic Regression showed around 80% accuracy, it completely failed at identifying fatal attacks, which were the cases we actually cared about most. That insight pushed the team to shift focus toward the Random Forest model, which performed much better in capturing those high-risk cases.

From there, I used the Random Forest's predicted probabilities to develop the project's risk score. Turning those probabilities into a single, interpretable number wasn't straightforward, and it involved a lot of trial and error, testing different approaches, and thinking carefully about how users would understand and apply the score. I had to balance technical correctness with simplicity, making sure the score was meaningful without being overly complicated. In the end, the risk score became one of the most impactful parts of the project because it translated complex model outputs into something practical and usable.

I also played a major role in integrating everything into the dashboard, which turned out to be one of the most frustrating and time-consuming parts of the project. Making sure the filters, charts, and risk calculator all worked together seamlessly required a lot of debugging and patience. Through this process, I gained a much deeper understanding of how front-end interactions connect with back-end logic, and how small issues can affect the overall user experience.

In addition, I handled the feature importance analysis, helping identify which variables had the biggest impact on the model's predictions. I also wrote the sections of the report that covered the experimental design and key findings, making sure the methodology was clearly explained and supported by the results. During the demo, I walked through the model performance and explained the dashboard visuals, especially the feature importance chart and the risk score distribution, helping the audience understand not just what the model predicted, but why.

Overall, my contributions were a key part of making the project both technically strong and practically useful. As discussed in the report, this project came with a lot of challenges, and one of the biggest strengths of my team was the collaboration we had. We consistently communicated the problems we were facing, worked through the issues together, and built on each other's ideas. Many of the best decisions came from those back-and-forth conversations, where one of us would catch something the other had missed or suggest a better approach. In the end, working on something we were both genuinely interested in made the process more meaningful, and we are truly proud of what we were able to build from start to finish.